Description of the Fair LiH Subspace Benchmark Experiment

P765 Final Project

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1 Purpose

This experiment compares three different basis-state families for quantum subspace methods on the LiH molecule:

- Krylov basis states,
- UCCSD basis states,
- fixed-architecture hardware-efficient ansatz (HEA) basis states.

The goal is to compare these three families fairly as *subspace bases*. Each family generates a small set of states, and those states define a projected Hamiltonian problem whose lowest eigenvalue approximates the ground-state energy.

2 Hamiltonian and Reference State

The LiH benchmark uses the PennyLane quantum chemistry Hamiltonian for the default LiH geometry already used in the project:

- molecule: LiH,
- bond length: 1.5474 Å,
- basis: `sto-3g`,
- unit: `angstrom`.

For this system, the notebook builds:

- the qubit Hamiltonian $\hat{H}$,
- the Hartree–Fock reference state,
- the electron number and qubit count,
- the exact FCI reference energy in the fixed-electron sector.

In the validated default run, LiH maps to 12 qubits, 4 electrons, and a Hamiltonian with 631 Pauli terms.

3 Basis Families

Krylov basis

Each Krylov state is generated from the Hartree–Fock reference using Trotterized time evolution:

$$|\phi(t)\rangle \approx e^{-it\hat{H}}|\phi_{\mathrm{HF}}\rangle.$$

One scalar time parameter is optimized for each added basis state.

UCCSD basis

Each UCCSD state starts from the same Hartree–Fock reference and uses PennyLane's `qml.UCCSD` template. For LiH, the default setup produces 92 trainable UCCSD parameters for each optimized basis state.

Fixed-architecture HEA basis

Each HEA state also starts from the Hartree–Fock reference. The circuit architecture is fixed globally:

- one layer,
- rotation gates `RY` and `RZ`,
- `CNOT` entanglers,
- chain connectivity.

Only the angle vector is optimized. For LiH, this gives 24 trainable HEA parameters per optimized basis state.

4 Common Subspace Objective

If the benchmark constructs basis states

$$\{|\phi_0\rangle, |\phi_1\rangle, \ldots, |\phi_{K-1}\rangle\},$$

then all families are compared using the same projected matrices:

$$H_{ij} = \langle\phi_i|\hat{H}|\phi_j\rangle, \qquad S_{ij} = \langle\phi_i|\phi_j\rangle.$$

The projected ground-state energy is the lowest generalized eigenvalue from

$$H\mathbf{c} = E_{\mathrm{sub}} S\mathbf{c}.$$

The same overlap-conditioning rule is used for all three families. In the current LiH package, a looser overlap cutoff is used so that nearly dependent states can still be retained when the generalized eigensolver remains numerically manageable. If the overlap matrix develops eigenvalues below the cutoff

$$10^{-10},$$

the candidate basis state is rejected and receives a large penalty. This prevents nearly duplicate states from unfairly enlarging the basis.

5 Fairness Protocol

The LiH benchmark follows the same fairness rules as the original H_2 notebook:

- same Hamiltonian,
- same Hartree–Fock reference state,
- same basis size target K,
- same projected-subspace objective,
- same optimizer family and budget,
- same overlap-conditioning rule.

The basis is built greedily:

1. the first basis state is the Hartree–Fock state,
2. one new basis state is optimized at a time,
3. previously selected basis states remain frozen,
4. the projected energy is recomputed after each addition.

6 Default LiH Notebook Configuration

The molecule-specific notebook `Fair_Subspace_Benchmark_LiH.ipynb` uses conservative defaults so the benchmark remains runnable:

- `mol_name = "LiH"`,
- `bond_length = 1.5474`,
- `basis = "sto-3g"`,
- `unit = "angstrom"`,
- `K_max = 3`,
- `device_name = "lightning.qubit"`,
- `optimizer_method = "Powell"`,
- `maxiter = 20`,
- `maxfev = 120`,
- `n_restarts = 2`.

The remaining fairness-related settings match the H_2 benchmark:

- `random_seed = 1234`,
- `t_max = 1.0`,
- `theta_max = 1.0`,
- `s_eval_cutoff = 1e-10`,
- one-layer fixed HEA with `RY`/`RZ` rotations and chain `CNOT` entanglers.

7 Default LiH Results

For the validated default run, the FCI reference energy is

$$E_{\mathrm{FCI}} = -7.882762250027 \text{ Ha}.$$

The projected subspace energies across the default sweep are:

Family	K	E_{sub} (Ha)	$E_{\mathrm{sub}} - E_{\mathrm{FCI}}$ (Ha)
Krylov	1	−7.863113895722	+0.019648354305
Krylov	2	−7.879582032481	+0.003180217546
Krylov	3	−7.881680528900	+0.001081721127
UCCSD	1	−7.863113895722	+0.019648354305
UCCSD	2	−7.863217813167	+0.019544436860
UCCSD	3	−7.863280641062	+0.019481608965
Fixed HEA	1	−7.863113895722	+0.019648354305
Fixed HEA	2	−7.863115166889	+0.019647083138
Fixed HEA	3	−7.863157332218	+0.019604917809

Under the same fair subspace protocol:

- Krylov gives the strongest improvement and now reaches the full default LiH sweep through $K = 3$ under the looser cutoff.
- UCCSD continues to improve from $K = 2$ to $K = 3$, but only slightly under the conservative budget.
- the fixed HEA also improves from $K = 2$ to $K = 3$, though the gain remains very small.

8 Interpretation

This LiH experiment answers the question:

> Given the same LiH Hamiltonian, the same Hartree–Fock starting point, the same subspace size, and the same optimization budget, which basis family gives the best projected ground-state energy?

For the conservative default settings used here, LiH remains a difficult optimization problem, but the looser cutoff makes the Krylov behavior much clearer: it can now add a third basis state and remains the strongest family across the full default sweep. UCCSD and the fixed HEA still improve only weakly with increasing K, so the LiH gap-vs-K trend stays much flatter than in H_4 for those parameterized bases.

9 Implementation Files

The LiH benchmark package is implemented in:

- `Fair_Subspace_Benchmark_LiH.ipynb`
- `Fair_Subspace_Benchmark_LiH_Description.tex`